Porch → Mudroom Conversion

Design, Engineering, Permits & Procurement Pack · Rev B

Private residence · Orange (84100), Vaucluse, France

SCOPE

Enclose the covered porch bay (≈ 2.0 × 4.0 m), raise the floor +0.30 → +1.25, and split it into an entry LANDING (2.0 × 1.4) behind new door K1 at the stair head + a fitted, insulated MUDROOM / vestiaire (2.0 × 2.5) behind door V

DELIVERABLES

Proposed plan · sections · elevation · two-door circulation & safety · method statement · French permits · joinery shop drawings (W1 · B1 · M1 · shoe tower) · FR vs TR cost & logistics

STATUS

Rev B — circulation revised (landing + K1 + V; old east door cancelled). Site measurement before fabrication is mandatory — no ordering or building from these nominal values.

Existing porch & stair · target mudroom concept · area to be converted

Project at a Glance

What is being built in Rev B, the four decisions that matter, and the headline numbers

THE PROJECT IN ONE PARAGRAPH (REV B)

The covered, open-sided bay next to the entrance stair (≈ 2.0 × 4.0 m internal, floor at ≈ +0.30 m) is enclosed with a new insulated masonry wall and its floor is raised ≈ 0.95 m to +1.25. Inside it is split in two: a 2.0 × 1.4 m entry LANDING behind a new glazed aluminium door K1 at the head of the stair (security + thermal buffer), and a 2.0 × 2.5 m fitted MUDROOM behind a light partition with timber door V. Walking in through V you face the mirror. Fit-out: wardrobe wall W1 (2 modules) on the west, bench B1 with hooks and lit niches on the east, mirror unit M1 on the north wall, shoe tower on the partition. The old east (stair-side) mudroom door of earlier drafts is cancelled — that wall is built solid.

FOUR DECISIONS THIS PACK RESOLVES

1 · CIRCULATION & THE TWO DOORS

Stair → K1 (opens INWARD onto the landing, hinge north — never out over the flight) → landing → V (opens INTO the mudroom, hinge west — the landing is too shallow for a swing). Old east door: cancelled, wall built solid. Details: pages 04–05.

2 · HOW TO RAISE THE FLOOR 0.95 m

Recommended: masonry sleeper walls + beam-and-block (poutrelles–hourdis) floor with a ventilated void, one level +1.25 across landing and mudroom — the standard French method. Alternative: compacted fill + slab (settlement risk). Details: page 06.

3 · PERMITS IN ORANGE

A Déclaration Préalable (Cerfa 13703) is mandatory: new façade wall + door K1 + high-level glazing band + closing a window. Orange has protected heritage zones — if in the ABF perimeter, add 1 month and follow render/colour guidance. Details: page 09.

4 · JOINERY: FRANCE vs TÜRKIYE

French bespoke: ≈ €9–14k installed. Turkish flat-pack custom production, shipped with A.TR (0% duty, 20% French import VAT) and self-assembled: ≈ €5.0–7.5k landed — typically 40–45% cheaper, with manageable risks. Details: pages 12–13.

HEADLINE NUMBERS (ORDER OF MAGNITUDE — REFINE AFTER SITE SURVEY & QUOTES)

≈ 8 m²
enclosed: landing 2.8 m² + mudroom 5.0 m² (verify)

+0.95 m
floor raise: +0.30 → +1.25 finished floor level

€8.4–13.7k
construction shell by artisan · €3.3–4.9k materials if self-build (excl. doors)

€5.0–7.5k
joinery landed via Türkiye route (vs €9–14k French bespoke)

1–2 mo
Déclaration Préalable processing (2 mo if ABF zone)

≈ €360–540
one-off taxe d'aménagement (8 m² × €892/m² × local rate)

WORKING ASSUMPTIONS — CONFIRM ON SITE BEFORE ANY ORDER (all values verified:false)

Internal bay 2.0 × 4.0 m and levels +0.30 / +1.25 come from the 2006 RDC plan and the owner's brief. The LANDING DEPTH OF 1.40 m IS AN ASSUMPTION — it cannot be read off the plan and is fixed on measure day; the glazing band (1000 × 450, sill +2000) is likewise an assumption. Soffit height (target ≥ 2.30 m clear after the raise), the window/hatch to close and the stair-head geometry must all be re-measured. Drawings are dimensioned for design intent, not fabrication — site measurement before fabrication is mandatory; joinery sizes are re-issued after the shell is complete.

Existing Conditions

Site photos, extract of the 2006 permit plan, and the level problem to solve

WHAT IS THERE TODAY

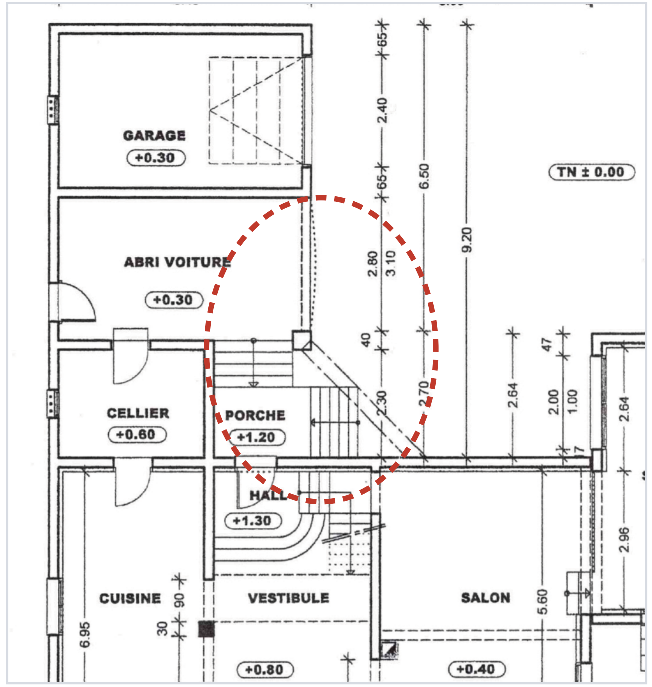
The entrance stair climbs ≈ 1.2 m from the driveway. Beside it, under the same roof, sits an open-sided covered bay at low level (≈ +0.30 m) currently housing the bins — this is the space to convert. The rear wall has a small barred window / hatch opening that will be closed. The bay is open to the driveway on one side; this opening receives the new insulated wall with door K1 at the stair head and becomes the main façade change.

1 · Entrance stair + open bay (new wall + K1 close this side)

2 · Inside the bay: low slab ≈ +0.30, hatch to close

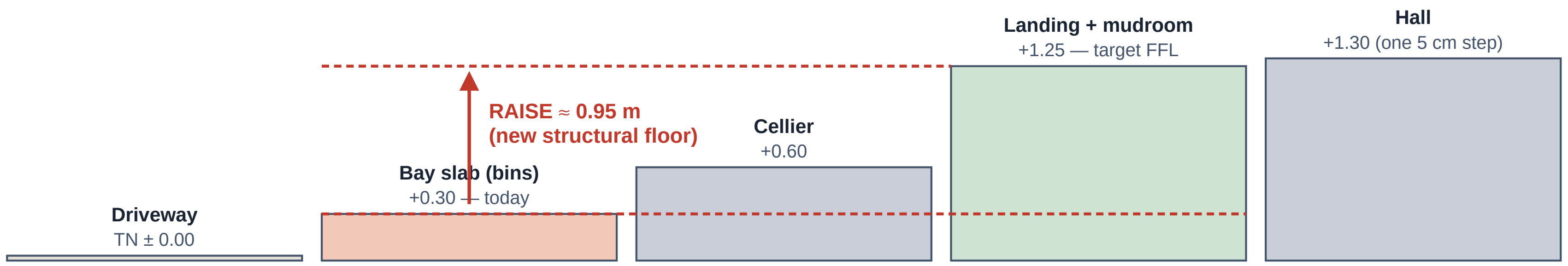
3 · Target: fitted mudroom (owner's reference)

Dashed red: conversion zone (indicative)



4 · RDC permit plan extract (2006) — porch zone

THE LEVEL PROBLEM — FIVE FLOOR LEVELS AROUND ONE CORNER



Key survey items before design freeze: ① exact bay clear width & length, ② soffit / roof timber height above +1.25 (need ≥ 2.30 m clear — photos suggest ≈ 2.4–2.6 m under the plate, verify), ③ LANDING DEPTH at the stair head — the 1.40 m in this pack is an assumption, ④ position & size of the window and hatch to close, ⑤ where roof rainwater discharges, ⑥ any services in the bay (the CCTV camera and cabling must be relocated).

Proposed Plan — Landing + Mudroom at +1.25

Rev B circulation: stair → K1 → landing → V → mudroom · red = new · grey = existing · blue = closed · dashed = joinery

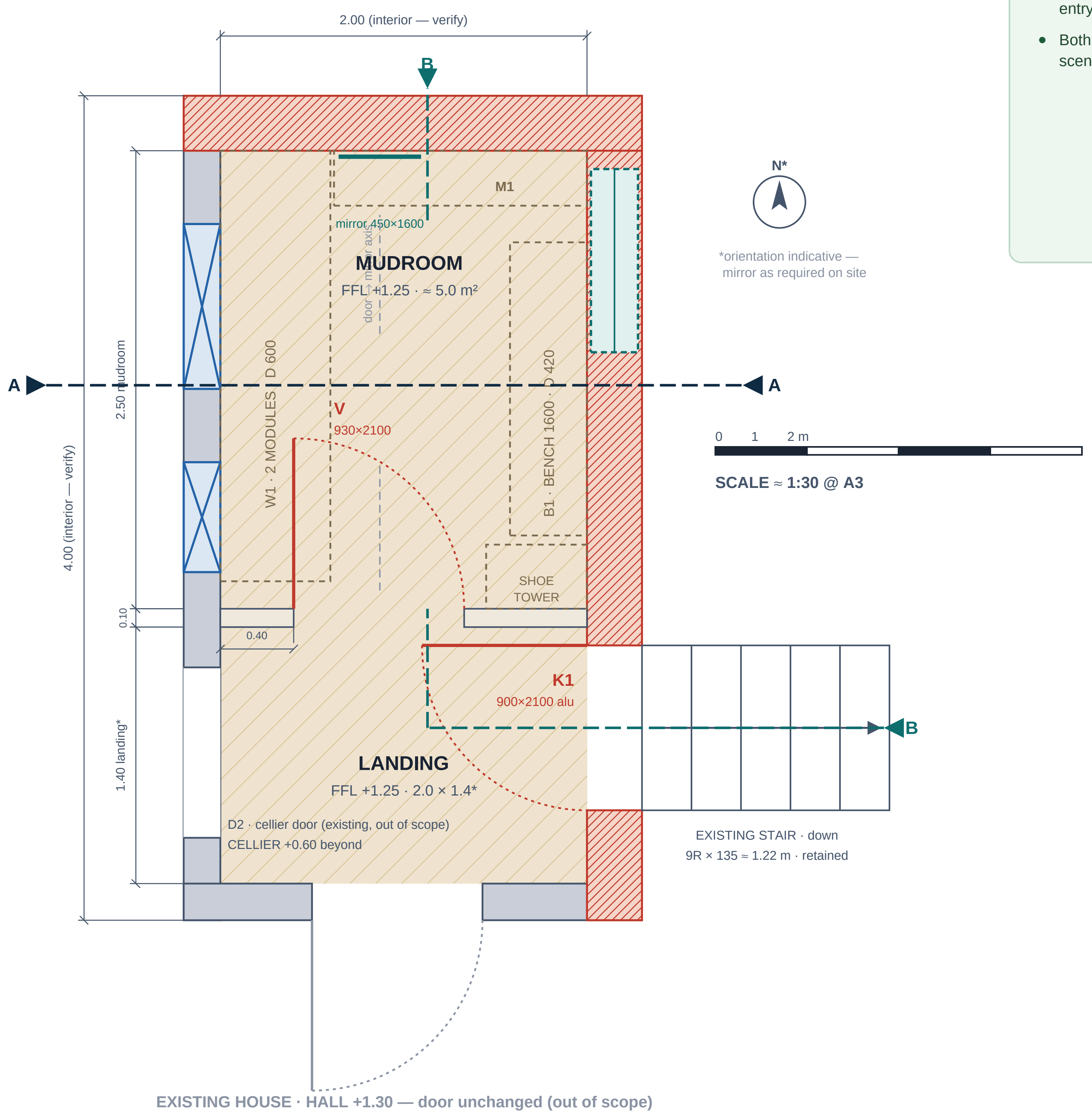
LEGEND

- NEW — insulated masonry wall (20 cm block + lining)
- EXISTING — retained walls
- Light partition — metal stud + 2×2 plasterboard, 100 mm
- CLOSED — existing window / hatch infilled
- Floor raised +0.30 → +1.25 (new structure)
- Fitted joinery (see shop drawings, p.10–11)
- Fixed glazing band 1000×450, sill +2000 (assumption)

PLAN NOTES

- N1 — K1: new lockable glazed aluminium door at the stair head. Opens INWARD onto the landing, hinge on the NORTH jamb; keep ≥ 90 cm of clear landing in front of the closed leaf.
- N2 — V: mudroom door in the light partition, 400 mm from the west wall. Opens INTO the mudroom, hinge WEST; entering, the mirror (M1) is straight ahead.
- N3 — New east & north walls on new strip footings (p.06); build tight to the soffit with a compressible joint. The east door of earlier drafts is CANCELLED — the east wall is solid except K1 and the glazing band.
- N4 — Window + hatch in the cellier wall infilled with blockwork (hatch may remain as an internal niche); W1 covers this wall, so internal finish is non-critical.
- N5 — Partition: 100 mm metal stud + 2 × 12.5 mm plasterboard each face + 45 mm mineral wool (sound); V frame fixed to doubled studs.
- N6 — Relocate CCTV camera and cabling before wall construction; provide 2 double sockets + LED feed for joinery (p.11).
- N7 — Floor finish: porcelain tile R10, landing and mudroom at the SAME level +1.25 — no step at V; K1 threshold ≤ 20 mm.

PROPOSED PLAN — REV B



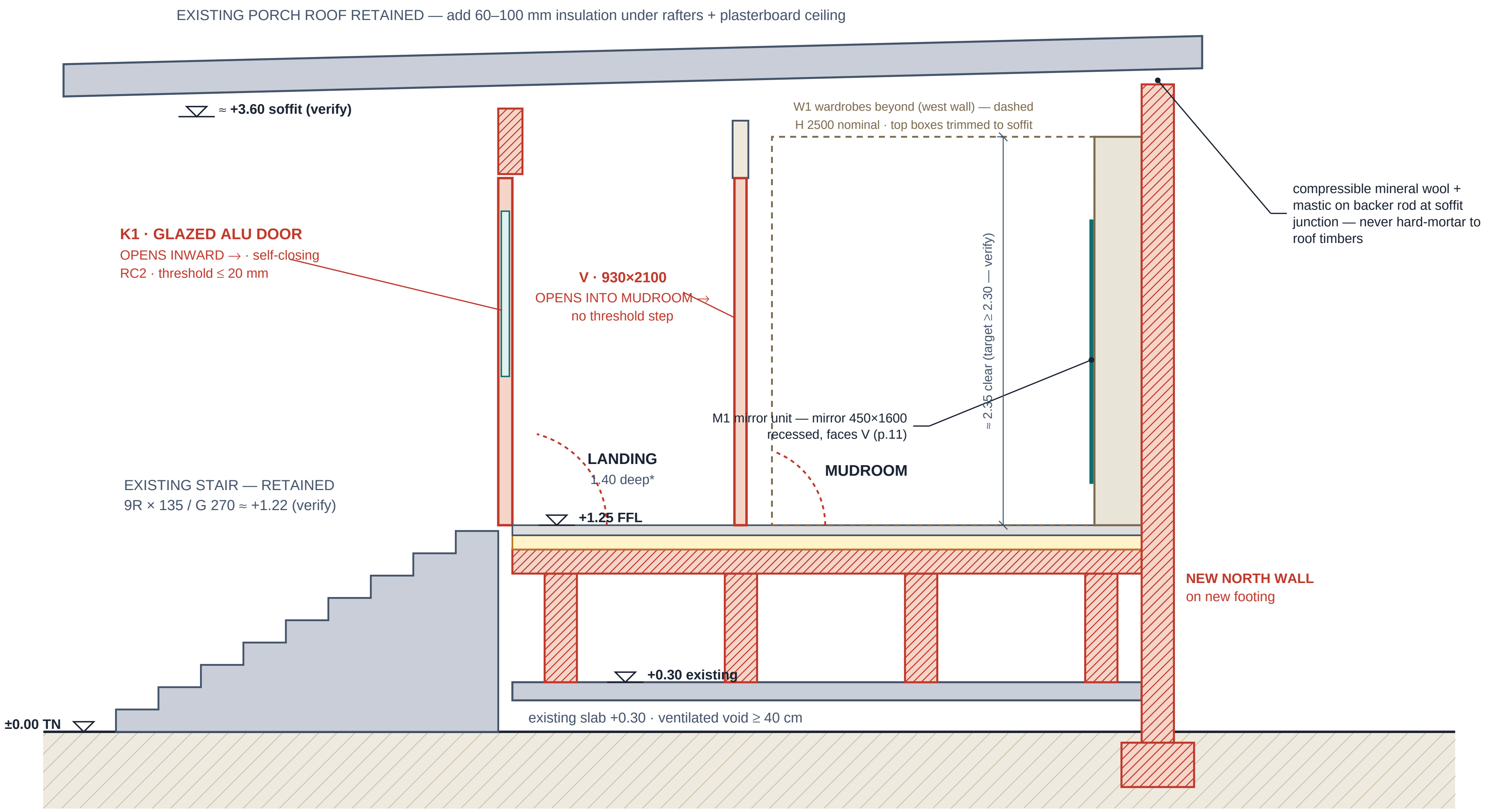
WHY THE DOORS SWING THIS WAY

design basis · dimensions.json · swing_rationale

- K1 — an outward leaf over the flight pushes the opener toward the stair void and can strike someone coming up: the classic fall scenario. K1 therefore opens INWARD onto the landing; self-closing hinges stop wind-slam. Hinge north: the open leaf parks against the landing's north wall, clear of the house door and the walking line.
- V — the landing is only 1.40 m deep, so V opens INTO the mudroom, hinge west: the leaf parks flat against W1's side panel and keeps the entry axis (door → mirror) clear.
- Both doors open at once = leaves in different rooms — no clash scenario. The existing house/hall door is untouched (out of scope).

Section B–B & The Two-Door Circulation

Developed cut along the entry path: up the stair → K1 → landing → V → mudroom → mirror



SECTION B–B · SCALE ≈ 1:35 @ A3 · developed along the entry path (up the stair, through K1, across the landing, through V, to the mirror)

*landing depth 1.40 m is an ASSUMPTION — fixed on measure day. Landing and mudroom floors are cast at the same +1.25 level: no step at V.

THE TWO DOORS: FINAL ANSWER

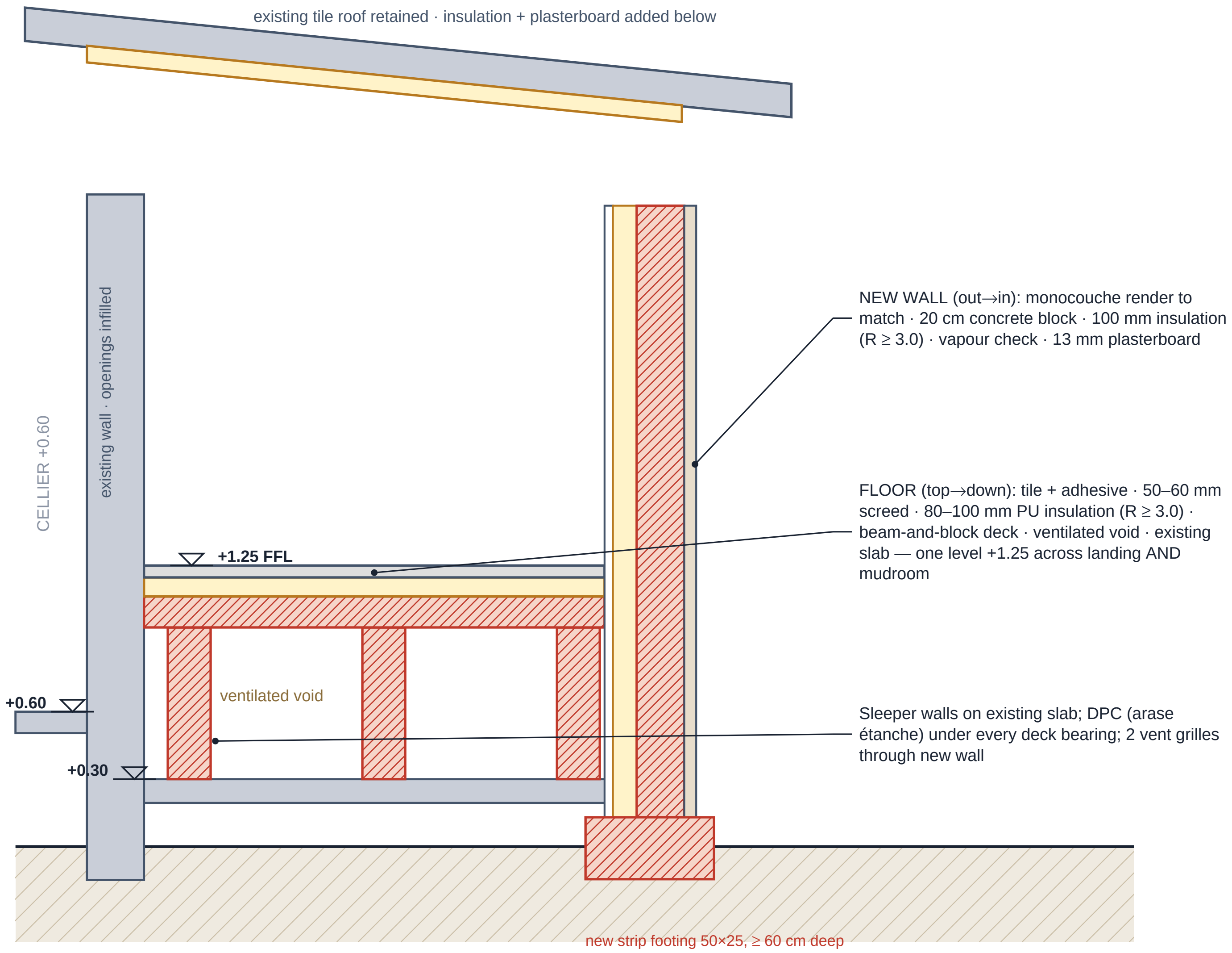
1. **Keep the stair exactly where it is.**
   Moving it means demolishing and recasting a sound concrete flight for zero functional gain. Not justified.
2. **K1 opens INWARD onto the landing — never outward over the flight.**
   An outward leaf forces the opener to stand on the steps and back down while pulling, and can strike someone coming up — the classic fall scenario. Self-closing hinges stop wind-slam.
3. **K1 hinge on the NORTH jamb.**
   The open leaf parks flat against the landing's north wall (the partition), clear of the house door and the walking line.
4. **V opens INTO the mudroom, hinge WEST.**
   The landing is only ≈ 1.40 m deep — a leaf swinging into it would sweep most of it. Hinged west, the leaf parks against W1's side panel and the entrant faces the mirror.
5. **Keep ≥ 90 cm of flat landing at K1 + the cheap safety extras.**
   Stand fully on the landing to unlock and step in — no foot on a tread. Continuous handrail, exterior sensor light, contrasting non-slip nosings, closer set slow.

DOOR SCHEDULE — K1 & V

- K1 — 900 × 2100 leaf, glazed thermal-break aluminium entrance door, Uw ≤ 1.6 W/m²K, RC2-class lock, self-closing hinges; façade change declared in the DP (p.09). Former draft code D3.
- K1 threshold: max 20 mm accessible aluminium thermal-break sill with drip; landing falls 1–1.5% away from the door.
- V — 930 × 2100 timber interior door in the partition, 400 mm from the west wall, Ud ≤ 1.4 W/m²K not required (both sides heated); hinge west, opens into the mudroom.
- V threshold: NONE — landing and mudroom are cast at the same +1.25 level (verify on site).
- Colour & style: K1 to match existing joinery colour — in an ABF zone the façade look must be approved with the DP (p.09).

Section A–A, Wall & Floor Build-Ups

Transverse cut through the mudroom: what every layer is, and the two ways to raise the floor 0.95 m



SECTION A–A · ≈ 1:25 @ A3 · looking north

RAISING THE FLOOR: TWO WORKABLE SYSTEMS

OPTION A — Beam-and-block on sleeper walls

RECOMMENDED · the standard French method (poutrelles–hourdis)

How
Build 3 low block walls on the existing slab, lay precast concrete beams + infill blocks across them, pour a thin topping, then insulation + screed + tile. A ventilated void (≥ 40 cm) remains below.

Why it wins
No settlement risk, light on the existing slab, keeps the floor dry (ventilated), fully DIY-able with two people, materials from any French builders' merchant (≈ €55–75/m² materials).

Watch-outs
Order beams cut to span (≈ 2.0 m — smallest standard section is fine); include 2 ventilation grilles through the new wall; DPC strip under every bearing.

VERDICT: use Option A unless the existing slab is unsound. ≈ €450–700 materials for 8 m².

OPTION B — Compacted fill + insulated slab

ACCEPTABLE · only with disciplined compaction

How
Fill the void with crushed stone (0/31.5) compacted in 20 cm layers with a plate compactor, blind with sand, lay polythene DPM + 100 mm PU insulation, cast a 10–12 cm mesh-reinforced slab, then tile.

Why consider
Conceptually simpler, no precast components to order, slightly cheaper on materials (≈ €45–60/m²), solid feel.

Watch-outs
≈ 6 m³ of fill to barrow in and compact; poor compaction = cracked, settling floor in year 1–2; adds ≈ 12 tonnes onto ground — confirm the bay slab/ground can take it or remove slab and fill from subgrade.

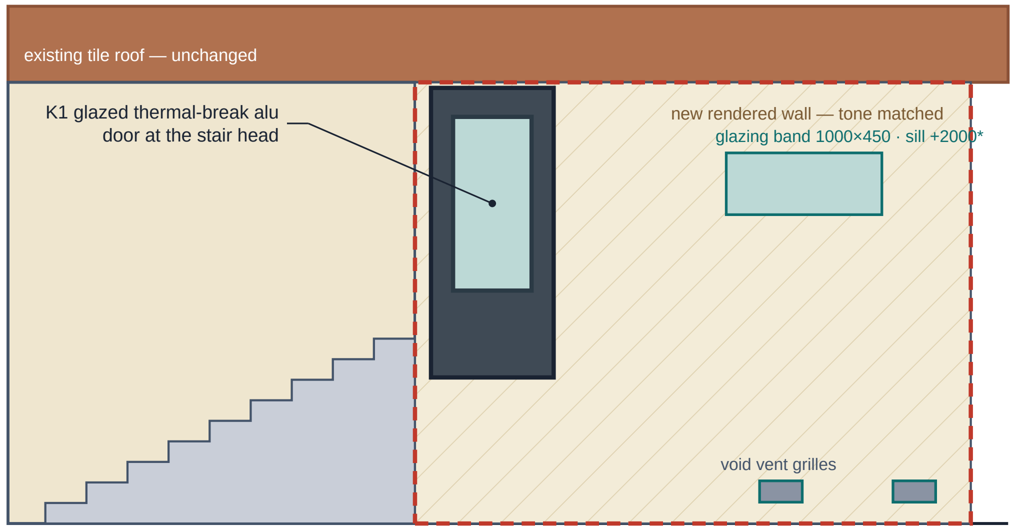
VERDICT: viable for a careful self-builder with a hired plate compactor; A remains safer.

Both options: lap the floor DPM/DPC into the wall damp-proof course, insulate the floor perimeter edge (thermal bridge), and cast landing and mudroom at the SAME +1.25 level — no step at door V; K1 keeps a ≤ 20 mm accessible threshold. The landing–mudroom partition separates two heated volumes: no thermal insulation required, 45 mm mineral wool for sound. Target envelope insulation: walls R ≥ 3.0, floor R ≥ 3.0, ceiling R ≥ 4.5 m²K/W — above the French “élément par élément” minimums (verify current arrêté values when buying).

Façade Elevation — Before / After

The only externally visible change: the open bay becomes a rendered wall with the new door

EXISTING · entrance façade · ≈ 1:60



PROPOSED · entrance façade · ≈ 1:60

EXTERNAL MATERIALS & WEATHERING DETAILS

NEW WALL

20 cm hollow concrete block (parpaing) laid in M10 mortar; every 2nd course tied to existing masonry with stainless wall starters (equerres) resin-fixed; bond beam (chainage) at top with 2×10 mm bars.

RENDER

Single-coat “enduit monocouche” (e.g. Weberpral M or ParexLanko), colour matched to the existing cream façade — take a chip to the merchant; finish “gratté fin” to match. In an ABF zone the exact tint may be imposed (p.09).

DAMP CONTROL

Bituminous DPC (arase étanche) on the footing at +0.15 above outside ground; render stops 10 cm above ground on a bead; gravel strip at the wall base; check the roof gutter discharges away from the new wall.

ROOF JUNCTION

Top of new wall meets the existing roof soffit: pack with compressible mineral wool, close both faces with mastic on backer rod — do NOT hard-mortar to the timber (movement).

CLOSING THE WINDOW

Remove frame & grille, tooth-in blockwork infill, insulate the reveal, render the outer face to match. Externally invisible when done well — still declared in the DP as a façade change.

HIGH-LEVEL GLAZING BAND

One fixed double-glazed band ≈ 1000 × 450 mm, sill +2000, in the east wall above the bench/niche zone (ASSUMPTION — confirm size & position on measure day). Daylights the mudroom without overlooking; K1’s glazing daylights the landing.

DECLARED FAÇADE CHANGES — BASIS FOR THE DP4 DRAWING

- Open bay enclosed with a rendered masonry wall (tone-matched monocouche)
- New glazed entrance door K1 at the stair head + high-level fixed glazing band 1000×450 (sill +2000, assumption)
- Barred window removed and opening infilled flush (rendered to match)
- Two low-level ventilation grilles for the underfloor void

Method Statement — Build Sequence for a Self-Builder

Fourteen steps in strict order; italic notes say what "good" looks like at each stage

1 Survey & mark-out

Re-measure the bay, levels (+0.30 / +1.25), soffit height, landing depth (1.40 m is an ASSUMPTION); mark wall + partition lines with chalk; photograph everything for the DP file.

2 Permission first

Submit the Déclaration Préalable and WAIT for approval (p.09). Display the site notice panel. No structural work before this.

3 Clear & protect

Relocate bins, trailer, mower; move the CCTV camera & cabling; protect the stair with boards; set up a dry materials area.

4 Footings

Excavate a 40 cm wide trench to ≥ 60 cm depth (frost) along the new wall lines; pour C25/30 strip footing 50×25 cm with 4×10 mm bars; cure 5–7 days.

5 Rising walls + DPC

Block up to slab bearing level; lay bituminous DPC on all new masonry at +0.15 above outside ground and on the sleeper walls.

6 Sleeper walls

On the existing +0.30 slab, build 3 low block walls (Option A) at ≤ 1.0 m centres under landing AND mudroom, level tops set for the beam soffit; leave 2 vent paths through to the new wall grilles.

7 Floor deck

Lay poutrelles + hourdis blocks, mesh + 4–5 cm structural topping — ONE level, +1.25 finished, across landing and mudroom; alternatively execute Option B fill & slab (p.06). Cure before loading.

8 Enclosure walls

Build the new east & north walls to the roof soffit — solid except the K1 opening and the glazing band (precast lintel over); cast the chainage bond beam; build in the K1 frame fixings.

9 Close the window

Strip frame & grille; tooth-in block infill; insulate reveal; leave outer face ready for render.

10 Roof junction & first fix

Pack and seal the wall/soffit joint; run electrics: 2 double sockets, ceiling light on sensor, 24 V LED feed for joinery, relocate camera (NF C 15-100: RCD-protected circuits).

11 Insulate, line & partition

Walls: 100 mm insulation + vapour check + 13 mm plasterboard. Ceiling: 60–100 mm + board. Floor: 80–100 mm PU + screed. Build the landing–mudroom partition: metal studs @600 + 2×12.5 BA13 each face + 45 mm wool; V frame on doubled studs.

12 External render

Monocouche render on the new wall + window infill, colour-matched, finish gratté fin; base bead 10 cm above ground.

13 Doors, floor finish, paint

Hang K1 (opens inward onto the landing, self-closer fitted) and V (opens into the mudroom); tile landing + mudroom at the same +1.25, R10 slip class; no step at V, ≤ 20 mm sill at K1; prime & paint.

14 Joinery & handover

Install W1 (2 modules), bench wall B1, mirror unit M1 and the shoe tower (pp.10–11); connect LEDs; silicone junctions; file the DAACT completion declaration at the mairie.

SITE SAFETY — NON-NEGOTIABLE FOR SELF-BUILD

Never undermine the existing stair or porch foundations when trenching — keep excavation ≥ 40 cm clear or underpin in short bays. Prop nothing off the roof timbers. Two people minimum for beam-and-block handling. Dust mask + eye protection when cutting block; gloves for mortar (cement burns). Hire, don't improvise: plate compactor, block cutter, laser level. If any crack appears in existing walls during works — stop and get a maçon to inspect.

French Permits & Rules — Orange (84100)

What must be declared, to whom, in what order — and what it costs

WHY THIS PROJECT NEEDS A DÉCLARATION PRÉALABLE (DP)

Three independent triggers apply, any one of which is enough: (1) enclosing a covered bay creates new "surface de plancher" — for 5–20 m² (up to 40 m² in a PLU urban zone) a DP is required, a full permis de construire is not; (2) any modification of the façade — the new wall, door K1, the glazing band and the closed window all count; (3) works in a heritage "abords" perimeter require authorisation whatever the size. One single DP file covers all of it. If the house total exceeds 150 m² after works, the DP is still valid (the 150 m² architect rule applies to permis de construire projects).

THE PROCESS, START TO FINISH

1. **Check the zone (free, 1 visit / email)**
   Ask the Service Urbanisme, Mairie d'Orange for: the PLU zone of the parcel, and whether it falls in the "abords des Monuments Historiques" / Site Patrimonial perimeter (Orange's Roman theatre & arch generate large protected radii). This determines render colour rules and adds ABF review time.
2. **Assemble the DP file (Cerfa n°13703)**
   DP1 site location plan · DP2 plot plan (masse) · DP3 section · DP4 façade before/after · DP5 roof plan if touched · DP6/DP7/DP8 photo-montage + near & far photos. The drawings in this pack (pp.04–07) are drafted to be adapted directly into DP3/DP4/DP6.
3. **Submit & display**
   File online via the town's guichet numérique or paper in 2–4 copies. Instruction: 1 month standard, 2 months in ABF perimeter (ABF silence = approval). Display the yellow site panel; third parties have 2 months to appeal from first display — safest to wait this out before building.
4. **Build, then declare completion**
   Works must match the approved file. On completion, file the DAACT (achievement & conformity declaration). Also update the property's taxable surface with the tax office (the DP data feeds this automatically).

MONEY & RULES ATTACHED TO THE DP

Taxe d'aménagement (one-off)
2026 base value €892/m² outside Île-de-France × ≈ 8 m² × (communal 1–5% + département ≤ 2.5%) ≈ €360–540, billed ≈ 12 months after approval.

Property tax
The extra 8 m² of habitable surface slightly increases taxe foncière — automatic, modest.

Thermal rules (RT existant, element-by-element)
Renovated/added elements must meet minimum R-values; this design targets walls & floor R ≥ 3.0, ceiling R ≥ 4.5, door Ud ≤ 1.4 — above minimums. Keep insulation invoices as evidence.

Insurance
Tell the home insurer: surface increase + works. For self-build, damage-ouvrage cover is not compulsory for an owner building for himself, but disclose the works.

Boundaries & neighbours
No new opening looks onto the neighbour closer than the legal view distances (1.90 m direct view); K1's glazing and the high fixed band face the own driveway — compliant. Confirm the new wall stays on own land.

■ ORANGE = MAJOR HERITAGE TOWN

The Théâtre Antique (UNESCO) and Arc de Triomphe create wide protected perimeters. If the parcel is inside one, the Architecte des Bâtiments de France reviews the DP (adds ~1 month) and can impose render tint, door colour and detailing. This is routine — design here already uses matching render and a sober door precisely to pass first time.

REALISTIC CALENDAR (SUBMIT IN SEPTEMBER → USE THE ROOM BY SPRING)

Prep DP file	Instruction 1–2 mo	Appeal window 2 mo (overlaps)	Shell works 3–4 wks	Fit-out + joinery 2–3 wks

≈ 5–6 months door-to-door;
joinery production runs in parallel

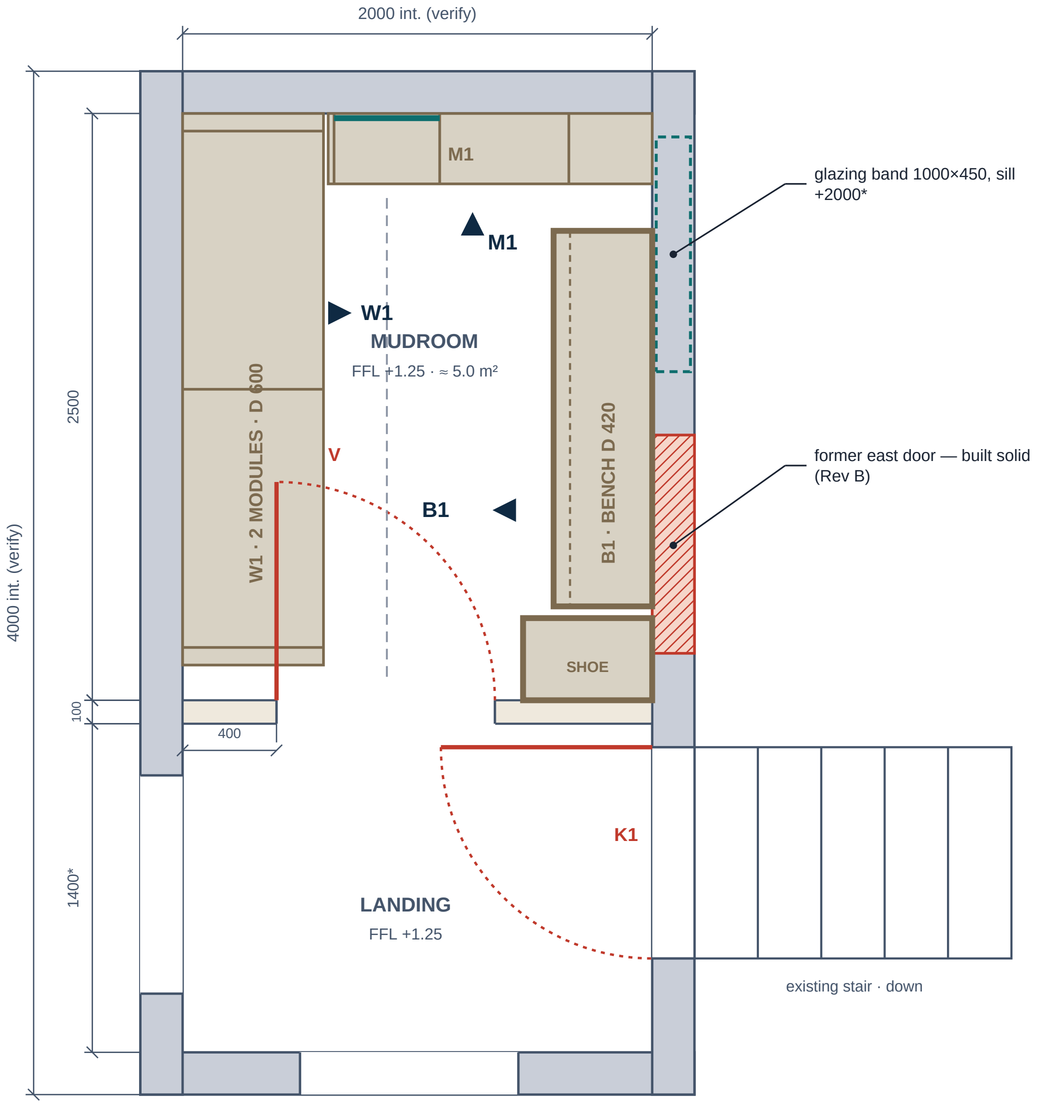
THE DP FILE — PIECE BY PIECE (CERFA N°13703)

- DP1 — site location plan (extract of cadastre / IGN)
- DP2 — plot plan (plan de masse) with the new wall dimensioned
- DP3 — section showing existing and new levels (adapt p.05–06)
- DP4 — façades before / after (adapt p.07)
- DP6–DP8 — photo-montage + near & far photos (use pack photos)
- Silence after the instruction period = tacit approval — keep proof of deposit

Joinery Shop Drawing 1 — Fitted Plan & Wardrobe Wall W1

The landing + mudroom fitted layout (Rev B), and elevation W1: two double-door wardrobe modules

FITTED PLAN — JOINERY LAYOUT (REV B)

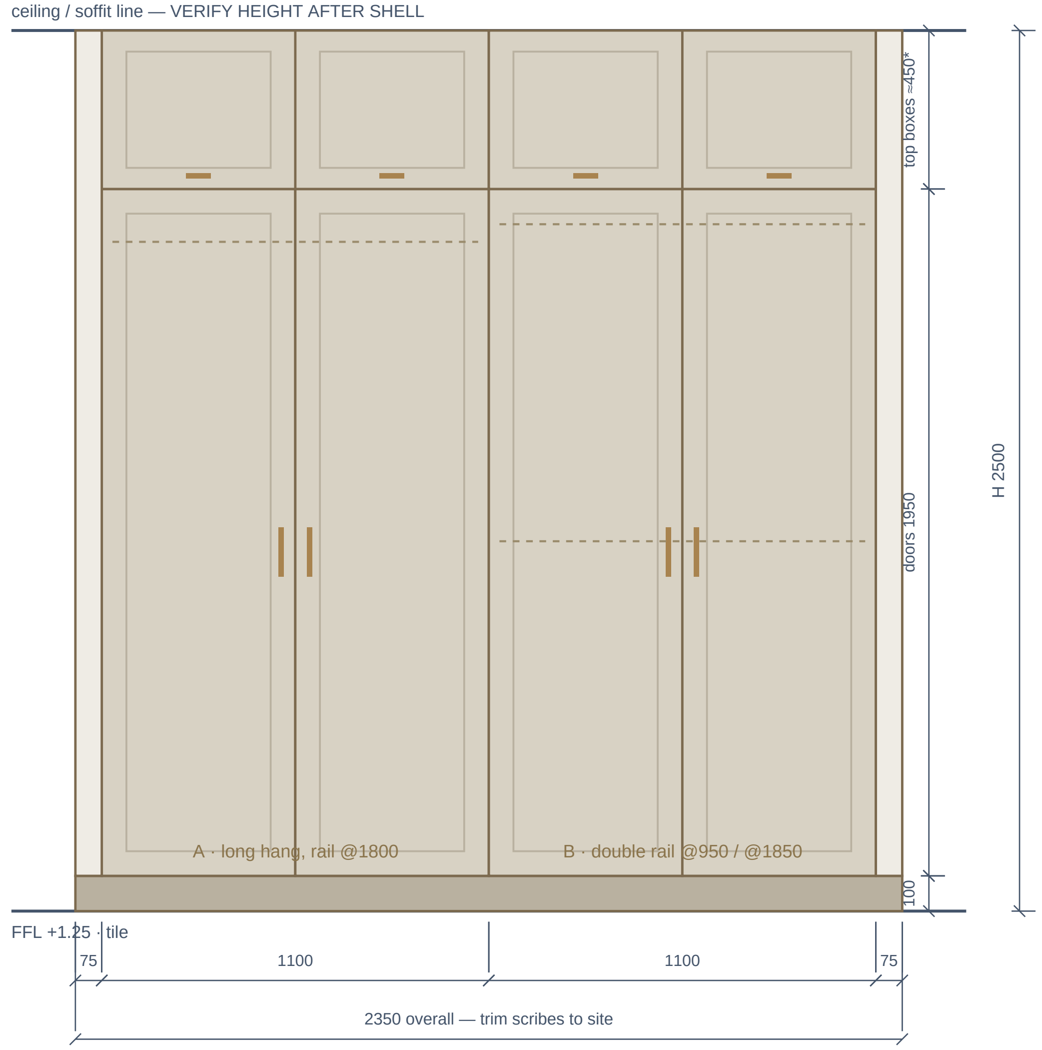


FITTED PLAN · ≈ 1:32 @ A3

clear aisle between W1 front and bench front ≈ 980 mm

*landing depth 1400 is an ASSUMPTION — measure day fixes it

ELEVATION W1 — WARDROBE WALL (LOOKING WEST)



ELEVATION W1 · ≈ 1:21 @ A3 — 2 double-door modules 1100 · doors 1950 + top boxes · depth 600

W1 NOTES (REV B)

- Rev B: the west wall run inside the mudroom is 2.50 m → 2 modules × 1100 + 75 scribes = 2350; the south 150 mm stays free — the V door leaf parks against W1's south side panel (felt stop, scratch-resistant lacquer there).
- Carcass 18 mm moisture-resistant MDF; doors 22 mm MDF shaker, PU-lacquered greige (≈ RAL 7044).
- *Top boxes are the tolerance zone: if the finished soffit lands below 2.60 m, shorten the top boxes — never the doors.
- Depth 600 external (≈ 560 internal) — full-size hangers fit. W1's north side panel is the abutment for the M1 mirror unit (p.11).
- Adjustable feet under plinth; scribe fillers 75 mm each end cut to the walls on site.
- Fix back rails into the blockwork with frame anchors — not into plasterboard alone.
- Site-verify: wall lengths, plumb (±15 mm absorbed by scribes), socket / LED feed positions before ordering.

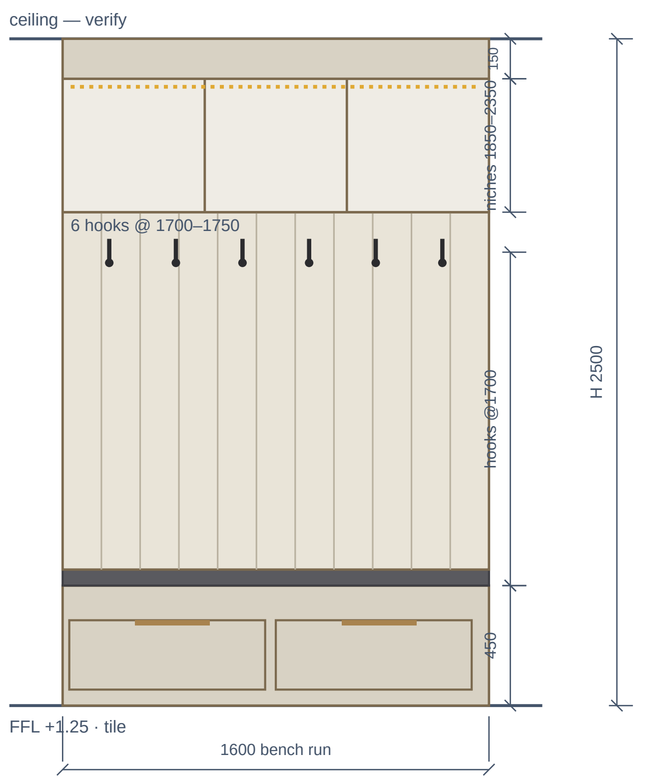
ORDERING RULE — SITE MEASUREMENT BEFORE FABRICATION IS MANDATORY.

These drawings fix the design intent. The fabrication cut-list is issued only after the room is built, tiled and re-measured — one visit, laser measure, update the elevations, then order.

Joinery Shop Drawing 2 — Bench B1, Mirror Unit M1, Shoe Tower

Elevation B1 (1600, 2 drawers) · north-wall mirror unit M1 on the V-door axis · shoe tower · sections · hardware

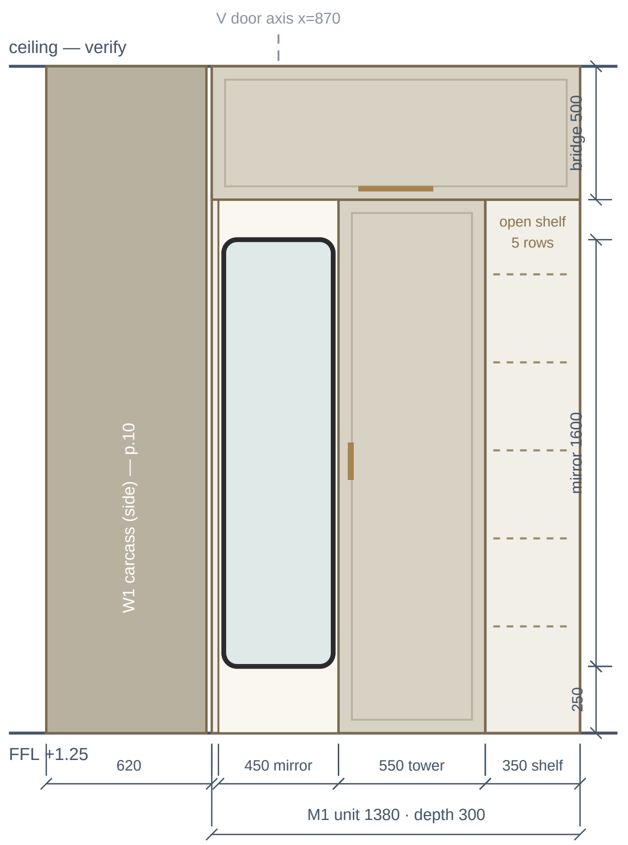
ELEVATION B1 — BENCH WALL (LOOKING EAST)



ELEVATION B1 · ≈ 1:28 @ A3

east wall — former door built solid, full wall usable

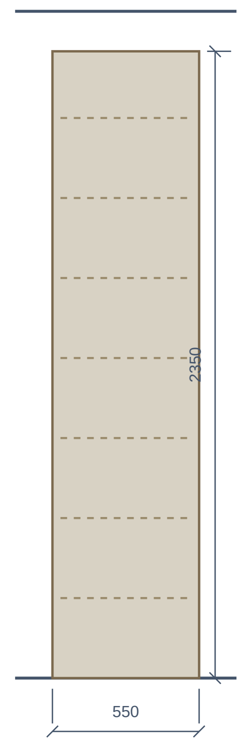
M1 MIRROR UNIT — NORTH WALL



M1 — NORTH WALL · ≈ 1:28 @ A3

mirror recessed on the V-door axis — the entrant faces it; west side abuts W1

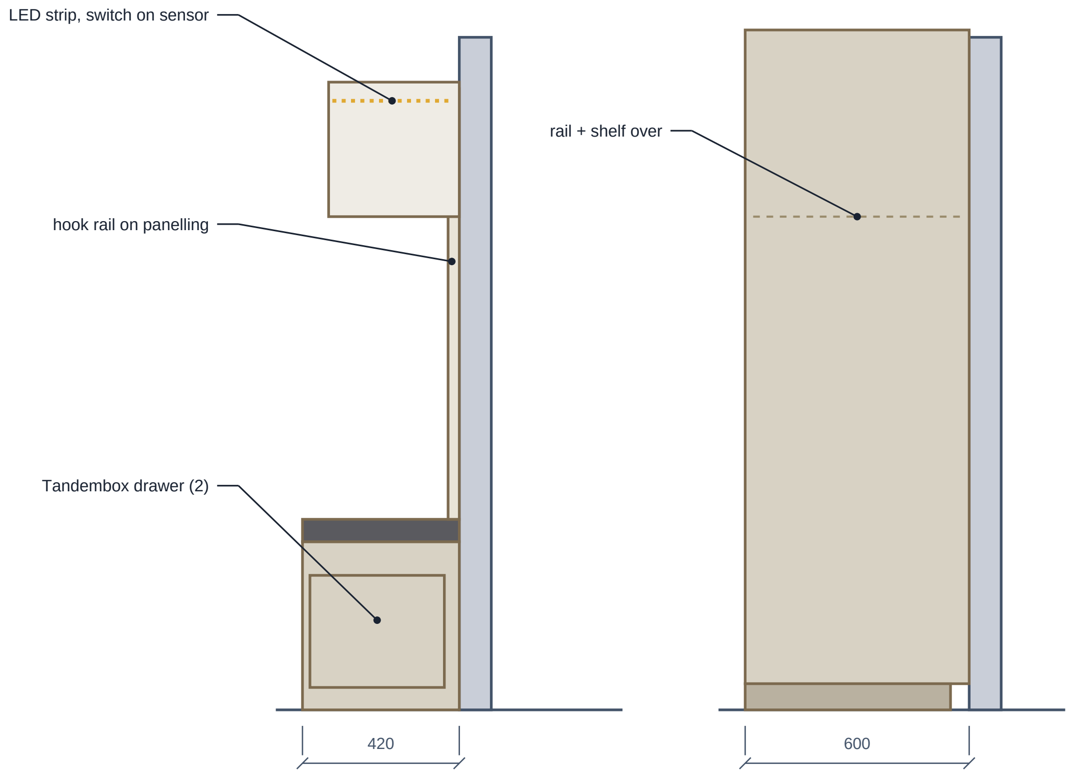
SHOE TOWER



SHOE TOWER · ≈ 1:28

550×350×2350 · ≈ 10 pairs
on the partition, east end
(anti-tip fixing only)

VERTICAL SECTIONS



THROUGH BENCH · 1:20

THROUGH W1 · 1:20

HARDWARE & MATERIALS — REV B COUNTS

Component	Specification
Carcass	18 mm moisture-resistant MDF, edged; backs 8 mm grooved-in
Doors & fronts	22 mm MDF shaker, PU-lacquered greige ≈ RAL 7044, matt
Hinges	Blum Clip-top 110° soft-close — 3 per tall door, 2 per top-box door (W1 total 20)
Drawers	Blum Tandembox, full extension, soft-close, 30 kg — 2 sets (bench B1)
Lighting	24 V 3000 K LED strip 2.0 m (B1 niche band); driver + sensor switch
Plinth & feet	100 mm plinth on 14 adjustable feet; clipped, removable
Scribes	75 mm scribe fillers (W1 ends) + ceiling scribes, site-cut
Pulls	Brushed brass or matt black, 128–160 mm centres — 10 (W1 8 + M1 tower + bridge)
Hooks	Matt black double hooks — 6, @ 1700–1750 on the B1 panel
Mirror	450 × 1600, black metal frame, recessed in M1 — 1

FABRICATION RULES — READ BEFORE ORDERING

- Site measurement before fabrication is mandatory — cut list only after the shell is complete and laser-measured.
- Walls will not be plumb: scribes absorb ±15 mm; never order joinery tight to nominal masonry sizes.
- The V leaf parks against W1's south side panel: felt stop + scratch-resistant lacquer there; keep that face clear of pulls.
- Seal all cut MDF edges (PU lacquer or edge tape) — this is a wet-shoes room.
- LEDs: run the 24 V feed before lining the wall (p.08, step 10); driver accessible above the niche top filler.
- If TR route (p.13): pack panels numbered per module (W1-A/B, B1, M1, ST shoe tower) with the assembly map.

Cost Plan — Shell & Joinery (Rev B)

Order-of-magnitude budget from the Rev B quantities: refine every line with quotes before committing

CONSTRUCTION SHELL (≈ 8 M²: LANDING + MUDROOM) — ORDER OF MAGNITUDE

Element	Artisan-built	Self-build (materials)
Groundworks & strip footings	€900–1,400	€400–550
Masonry walls + external render (≈325 blocks, former door built solid)	€2,200–3,400	€1,100–1,500
Floor raise 0.95 m — Option A (beam & block)	€1,000–1,600	€450–700
Window / hatch infill	€400–700	€150–250
Landing–mudroom partition (studs + 2×2 BA13 + wool)	€400–700	€150–250
Roof junction + ceiling insulation & board	€700–1,100	€300–450
Electrics (NF C 15-100, incl. camera move)	€600–900	€250–400
Door K1 — glazed thermal-break alu, RC2 (fit / supply)	€900–1,600	€700–1,200
Door V — interior timber 930×2100	€400–700	€250–450
Glazing band 1000×450 fixed (assumption)	€300–500	€200–350
Tiling + decoration	€600–1,100	€250–400
TOTAL	≈ €8,400–13,700	≈ €4,200–6,500 incl. doors

Self-build materials excluding doors K1 + V ≈ €3,250–4,850 — consistent with the €3.3–4.9k headline on p.02. Quantities from data/quantities.json (Rev B): ≈325 blocks, 28 bags mortar, ≈1.5 m³ concrete, 8 poutrelles + 8 m² hourdis, 12.2 m² partition board, 20.9 m² wall insulation, 8.8 m² tile. Assumes a sound existing slab. Add 10–15 % contingency to whichever route you choose.

JOINERY — THREE ROUTES FOR THE REV B FIT-OUT (W1 + B1 + M1 + TOWER)

Route	Landed cost	What you get / watch-outs
FR bespoke menuisier, installed	€9,000–14,000	Painted MDF; full warranty; zero logistics
FR semi-custom (IKEA PAX/BESTÅ + custom fronts)	€3,500–6,000	Middle path; fast; DIY-friendly; look is near-reference, not exact (no recessed mirror unit)
TR custom flat-pack (CNC cam+dowel), shipped	€5,000–7,500	Ex-works €2,800–4,800 + freight €900–1,500 + 20 % import VAT + clearance €150–300 (+ local fitter €500–900 optional)

RECOMMENDATION

TR route ≈ 40–45 % cheaper than FR bespoke for the reference look — choose it only with a trusted atelier and final measures taken after the shell. Otherwise FR semi-custom is the pragmatic middle.

ALL-IN SCENARIOS — SHELL + DOORS + JOINERY (EXCL. TAXE D'AMÉNAGEMENT €360–540, P.09)

≈ €7.7–12.5k
Self-build shell + FR semi-custom
Cheapest overall — look is approximate

≈ €9.2–14k
Self-build shell + TR custom
Reference look at the lowest cost — most owner effort

≈ €13.4–21.2k
Artisan shell + TR custom
Least owner labour for the reference look

≈ €17.4–27.7k
Artisan shell + FR bespoke
Turn-key, full warranty — premium price

BUDGET DISCIPLINE — THREE RULES

1 · Never order joinery (or door K1) before the shell exists and the landing depth is measured — measurement error is the #1 cost risk of the TR route. 2 · Get two artisan quotes even if self-building: they price the risk you are taking on. 3 · Fix the K1, glazing-band and tile choices before the DP file — changing the façade after approval means a new DP.

Türkiye → France — Customs & Logistics

How the joinery travels: duty-free under A.TR, 20 % French import VAT, 5–8 days on the road

THE ROUTE, DOOR TO DOOR

EU–TR CUSTOMS UNION: FURNITURE = 0 % DUTY WITH A.TR

BUDGET THE 20 % FRENCH IMPORT VAT — IT IS NOT OPTIONAL

1 · ATELIER — ISTANBUL
CNC production 3–5 weeks after final site measures; QC photos before wrapping

→

2 · EXPORT + A.TR
Exporter / forwarder issues the A.TR movement certificate (valid 4 months)

→

3 · CMR ROAD GROUPAGE
Shared truck via the Balkans corridor; 5–8 days transit; CMR consignment note

→

4 · FRENCH IMPORT
0 % duty with A.TR · 20 % VAT on goods + freight · clearance fee €150–300

→

5 · DELIVERY — ORANGE
DAP: kerbside delivery; check panels against packing list; note damage on CMR

CUSTOMS MECHANICS — WHAT ACTUALLY HAPPENS

- EU–Türkiye Customs Union: industrial goods incl. furniture enter duty-free with an A.TR movement certificate issued via the exporter. No A.TR = tariff exposure — make it a contract condition.
- French import VAT of 20 % is due on goods value + freight at import; the forwarder advances and re-invoices it with a clearance fee ≈ €150–300.
- Documents: commercial invoice · packing list · CMR consignment note · A.TR. Keep copies; VAT receipt supports resale value and insurance.
- Buy DAP (Delivered At Place) with customs clearance included — one price, one responsible party, no surprises at the border.
- Private individual importing for own use: normally outside commercial-operator obligations (e.g. EUDR) — have the forwarder confirm in writing.

PACKING SPECIFICATION — MAKE IT IN THE CONTRACT

- Flat-pack CNC panels, cam + dowel joinery — no glued frames; every panel numbered (W1-A1, B1-D2, M1-K3, ST …) matching an assembly map PDF.
- Edges: solid protectors on all painted fronts; panels face-to-face with foam interleaf; moisture-barrier wrap; export-grade pallets or crates. The mirror travels in a dedicated crate.
- Max single panel weight two-person safe (≈ 40 kg); hardware bagged per module + 5 % spares; hinges & runners in original Blum boxes.
- Photograph every pallet before wrapping and after loading — the only accepted evidence for a transit-damage claim.
- Insure the shipment for replacement value incl. VAT & freight, not ex-works price.

ON ARRIVAL — THREE WAYS TO ASSEMBLE

SELF-ASSEMBLE — **€0 extra**
Cam + dowel = IKEA-level difficulty at larger scale. Two people, 2–3 days, laser level, patience with scribes. Best if you built the shell yourself.

TR TEAM TRAVELS — **≈ €800–1,500**
Flights + board for two fitters, 2 days. Fastest and they know the product — but agree liability, insurance and snagging in writing before booking.

LOCAL FITTER (RECOMMENDED) — **≈ €500–900**
A local menuisier-poseur for 2 days once goods clear customs. Balanced cost, French-speaking accountability, can also scribe and seal. Book ahead.

Recommendation, Timeline & Risk Register

What to do, in what order, what can go wrong — and the render prompt for the 3D view

THE RECOMMENDATION IN FIVE LINES

1. **Enclose, raise and split — it is the right project.**
   Small scope, big daily payoff: a secure buffered entry + a warm fitted mudroom replacing a bin bay. Budget honestly: ≈ €9.2–14k all-in self-build + TR joinery, ≈ €17.4–27.7k turn-key (p.12).
2. **Floor: Option A (beam-and-block on sleeper walls), one level +1.25.**
   No settlement risk, ventilated and dry, self-build friendly; landing and mudroom cast together — no step at V. Option B only with a hired plate compactor and discipline.
3. **Doors: K1 inward onto the landing (hinge north) · V into the mudroom (hinge west).**
   Settled on p.05 — no door ever swings out over the flight; the V leaf parks on W1's side panel and the entrant faces the mirror. Self-closer on K1, handrail, sensor light.
4. **Permits: DP first, always.**
   Submit the Déclaration Préalable, wait out instruction (+ ABF month if applicable) and the appeal window before structural work. p.09 is the checklist.
5. **Joinery: TR custom if the atelier is trusted — measured after the shell.**
   Otherwise FR semi-custom. Never order before the room exists. Logistics recipe on p.13.

MASTER TIMELINE (SUBMIT DP IN SEPTEMBER)

Sep · survey + DP file	Oct–Nov · instruction + ABF	Nov–Dec · appeal window	Dec–Jan · shell 3–4 wks	Feb · fit-out 2–3 wks

Joinery: measure after shell → produce → ship → install (parallel lane)

Room in use by early spring — ≈ 5–6 months door-to-door, of which ≈ 3 months is waiting for paperwork, not building.

RISK REGISTER

Risk	Likelihood	Impact	Mitigation
Measurement error (joinery ordered off-plan)	Medium	High	Iron rule: cut list only after shell is built and laser-measured (p.10–11)
Landing depth assumption (1.40 m) proves wrong	Medium	Medium	Measure day fixes it before any order; K1, partition and joinery re-issued from dimensions.json — do not improvise
DP refused / ABF imposes conditions	Low–Med	Medium	Pre-check zone at mairie; matching render + sober door designed to pass; adjust tint if asked
Settlement (only if Option B floor)	Medium	Medium	Prefer Option A; if B: 20 cm compacted layers, hired plate compactor, mesh slab
Transit damage (TR route)	Medium	Medium	Packing spec in contract, photos before wrap, insure landed value, note damage on CMR at delivery
Budget creep	Medium	Medium	10–15 % contingency; two artisan quotes; fix façade choices before DP

End of pack · Rev B · July 2026 — site measurement before fabrication is mandatory; not for construction until the DP is approved.

APPENDIX — 3D RENDER PROMPT (READY TO PASTE)

Paste into any image model to visualise the finished Rev B room:

"Photorealistic interior render, compact mudroom 2 m wide and 2.5 m deep, entered through a timber door in the short wall; floor-to-ceiling greige shaker wardrobes with brass pulls along the entire left wall; on the right an upholstered charcoal bench over two drawers, panelled wall behind with black coat hooks at 1.7 m holding beige and navy coats, three open niches above with warm 3000 K LED strip lighting and wicker baskets; facing the door a tall black-framed mirror recessed into a greige cabinet wall with a closed tower and open corner shelves; slim shoe cabinet beside the door; large beige porcelain floor tiles with a woven jute runner; soft daylight from a high slot window, 24 mm lens, eye level, architectural photography style"

Tip: render once per option — brass vs matt-black pulls — and let the household vote before ordering hardware.

NEXT THREE ACTIONS

- Measure the bay: width, length, soffit height, LANDING DEPTH (checklist p.03).
- One email to Service Urbanisme, Mairie d'Orange: PLU zone + ABF status of the parcel.
- Assemble and submit the DP file (checklist p.09) — the clock only starts when it is filed.